QUANTUM ARENA 1.0

Disaster Alert & Evacuation Planning System

- **Problem Statement Title-** An **AI-powered Forest Fire + Tsunami Defence Ecosystem** – from early detection to evacuation, response, and recovery.
- **Theme -** Smart City
- **PS Category -** Software
- **Team Name –** NeuroNix

Forest Fire

IDEA TITLE

We propose a ***Geospatial AI/ML + Quantum*** system for Bharat that predicts next-day forest fire risk zones, simulates hour-wise fire spread, and enables **real-time alerts, live monitoring,** and **smart evacuation planning** for **faster response** and effective disaster management.

Datasets

Satellite: MODIS, Sentinel-2

Weather: ERA5 (ECMWF)

Vegetation: NDVI from Sentinel-2

Human Activity: GHSL, OSM

Terrain: SRTM DEM (NASA)

AI + Simulation + Alerts

Fire Risk Prediction

ConvLSTM + UNet Hybrid Model

Fire Spread Simulation

CA-based Fire Spread Model

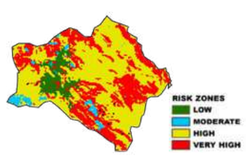
NDVI Delta + Sentinel-2 Imagery + BAI Index

Burned Area Estimation

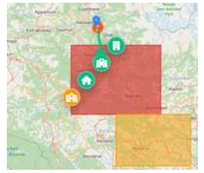
Automated Alert Engine + GIS Viewer

Alerting & Mitigation

"Export GeoTIFF / Shapefiles for Govt use"

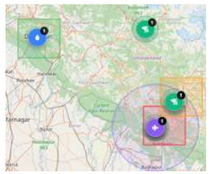


Predicted Fire Risk (U-Net Output)

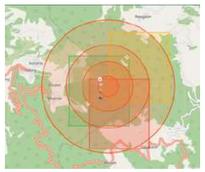


Safe Evacuation Route Mapping

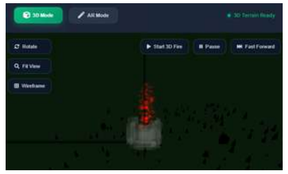
Carbon Emissions & Environmental Impact



Resource Optimization Engine

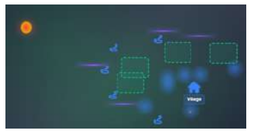


AI Explainability & Trust Layer



3D/AR Fire Visualization

Fire Spread Simulation of High Risk Zones (CA)



Community & Volunteer Engagement

Alert System

Figure 1. Brief idea about solution flow

Forest Fire

TECHNICAL APPROACH

QUANTUM ARENA 1.0

Multi-Source Data Ingestion

Satellite: MODIS, Sentinel-2

Weather: ERA5 (ECMWF)

Terrain: SRTM DEM (NASA)

Human Activity: GHSL, OSM

Vegetation: NDVI from Sentinel-2

Preprocessing & Feature Engineering

Normalize, Resampling, align datasets

Extract -
NDVI,
slope,
wind speed,
fuel moisture.

Grid everything to **30m resolution**

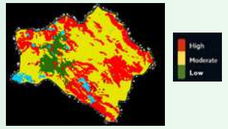
Combine All data

structured input tensor

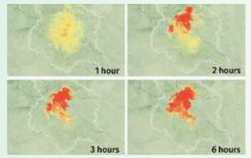
ML/DL

ConvLSTM + U-Net Hybrid Model

Accuracy **97%**



GIS-Ready Fire Risk Map (GeoTIFF) -Next Day



Fire Spread Simulation (CA-based) -1hr, 2hr, 3hr, 4hr, 6hr, 8hr, 12hr

Quantum Dispatch Engine

- QAOA-based optimization
- Allocates firefighters, drones, helicopters and water tanks
- Optimal deployment locations

AI Autonomous Alert Engine

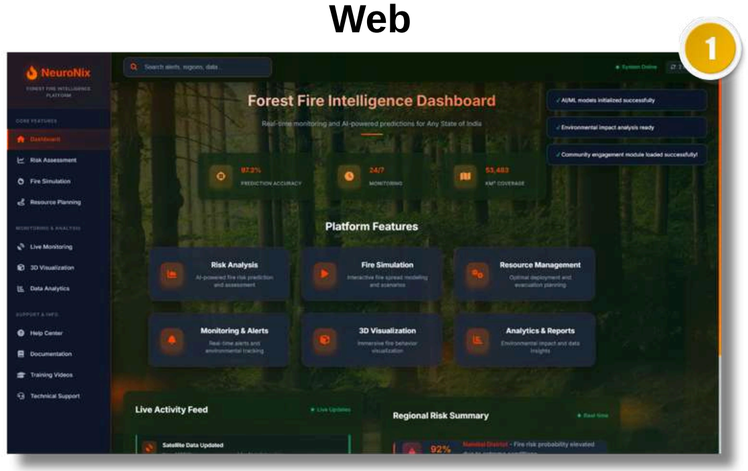
- Event-driven AI agent
- Triggers when fire risk crosses threshold
- Sends SMS alerts using Twilio API
- Integrates with GIS dashboard

1.**GeoTIFF, PNG, API-ready**
2.**District-level alerts**
3.Integrates with **ISRO Bhuvan /State forest sys*ems**

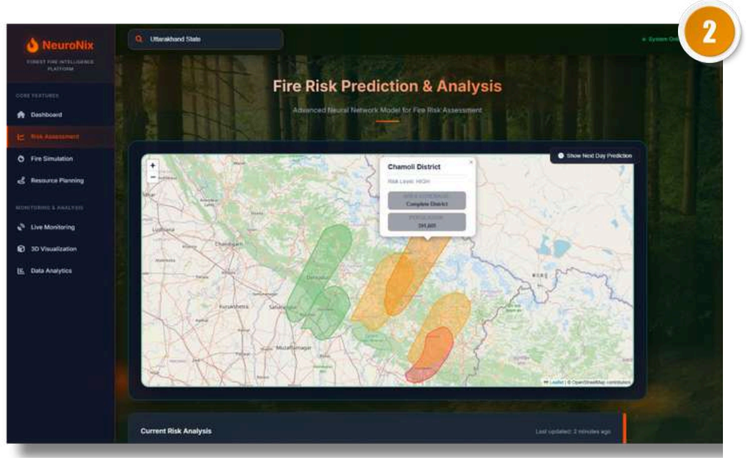
Fire Risk Prediction → Fire Spread Simulation → Quantum Evacuation Optimization → AI Alert Agent → SMS Alerts + Evacuation Routes

Figure 2. Architecture Diagram

Web



App

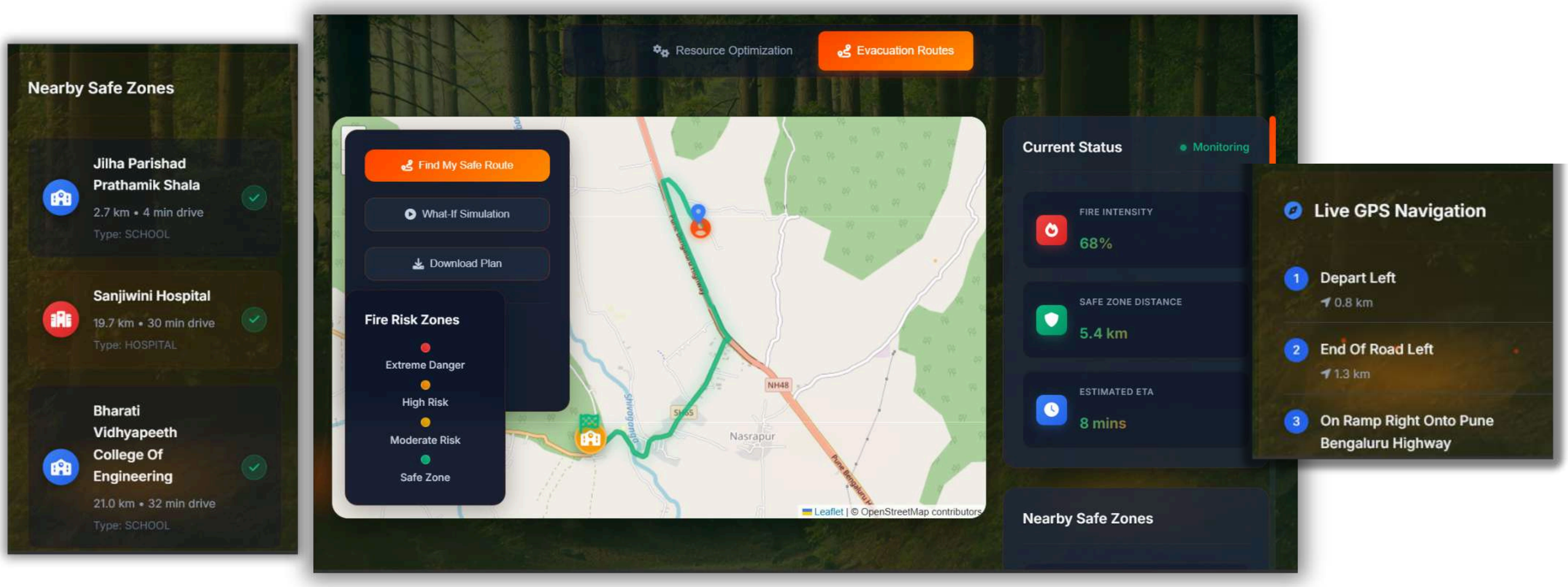


Mock diagrams of the proposed solution

Forest Fire

Precision Evacuation & Life Safety

*Navigating the **safest path** in **real-time** when seconds matter.*

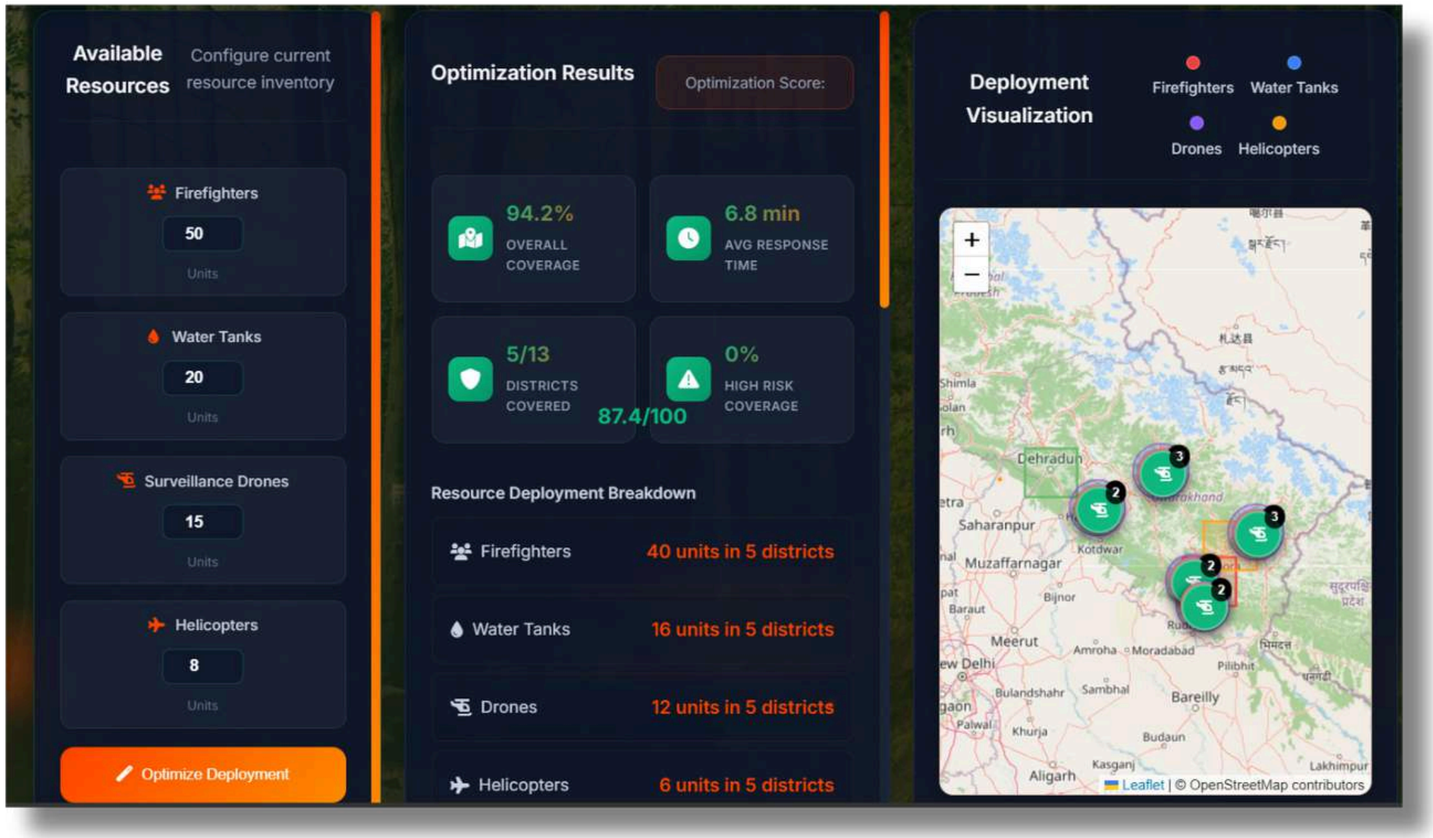


- Smart Route Optimization
- Instant AI Safe Zones
- Live GPS Emergency Navigation (Real-Time)
- Predicts risks and road closures in advance

Forest Fire

The Quantum Advantage in Crisis Management

*Moving from Classical approximation to **Quantum Computing** in Resource Allocation.*



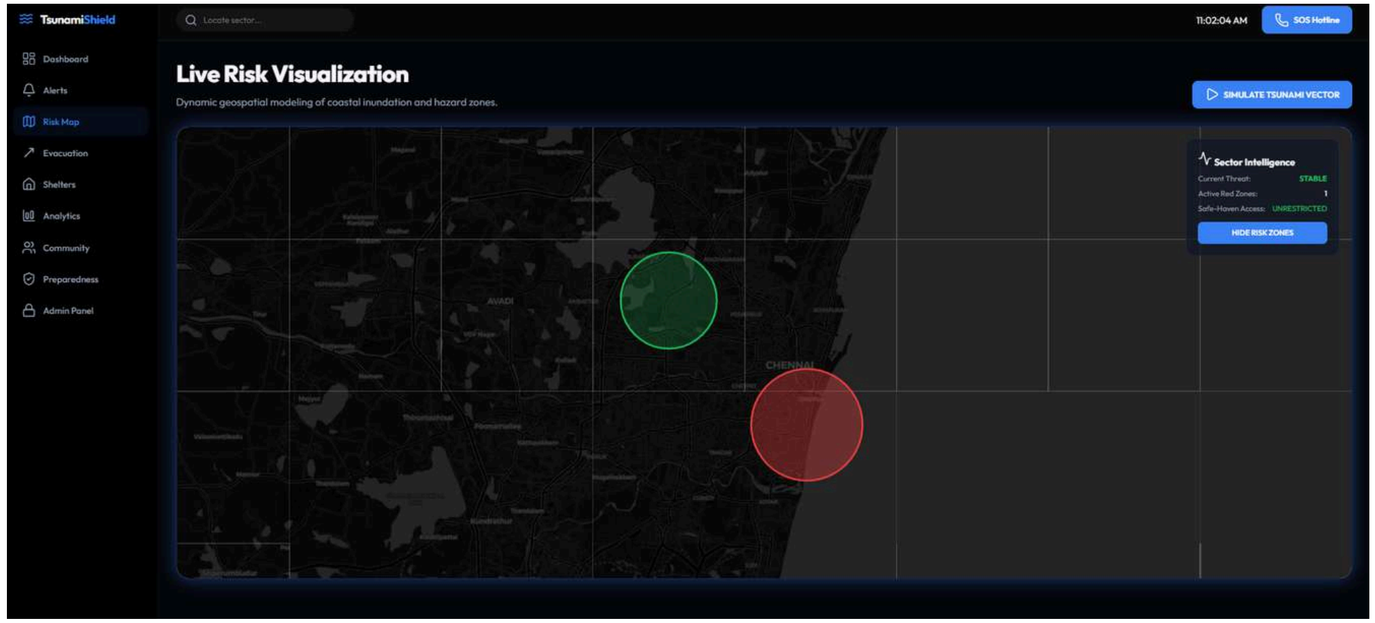
Performance Metrics (The "Proof")

- **68.3× faster decisions** compared to traditional CPU-based methods
- Ultra-low latency: **3ms (quantum) vs 200ms (classical)**
- **94.2% coverage**, ensuring nearly all high-risk zones are protected

"Harnessing the laws of physics to protect our forests. Tomorrow's defense, deployed today."

Tsunami

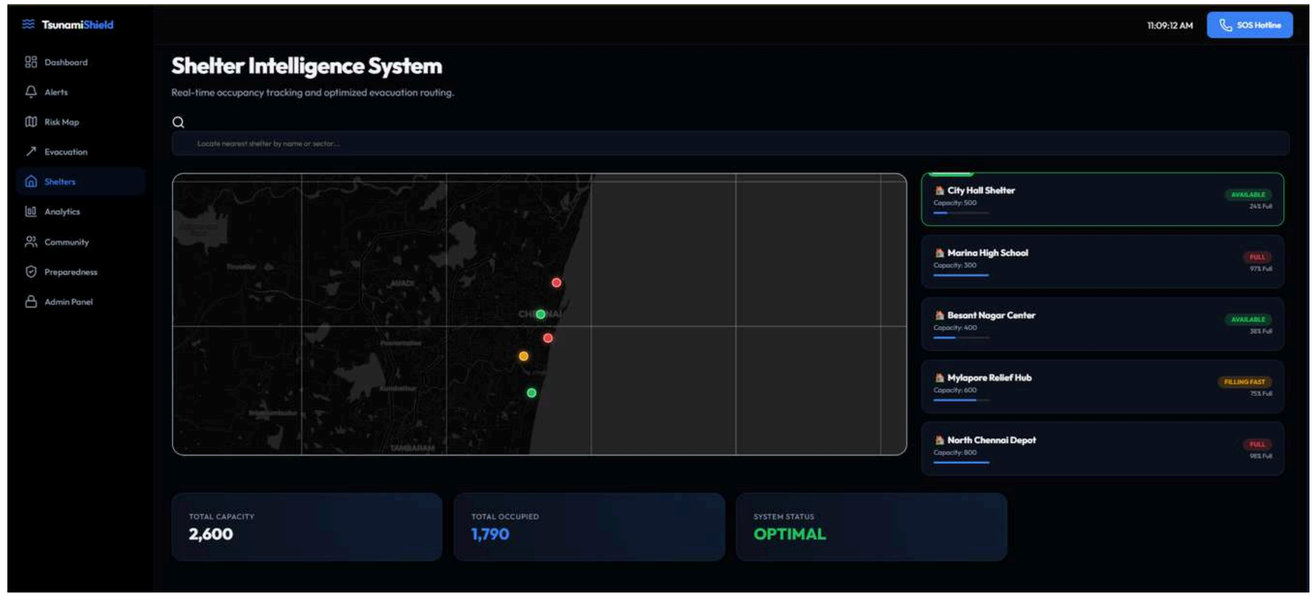
UI OF TSUNAMISHIELD



Shelter Intelligence System

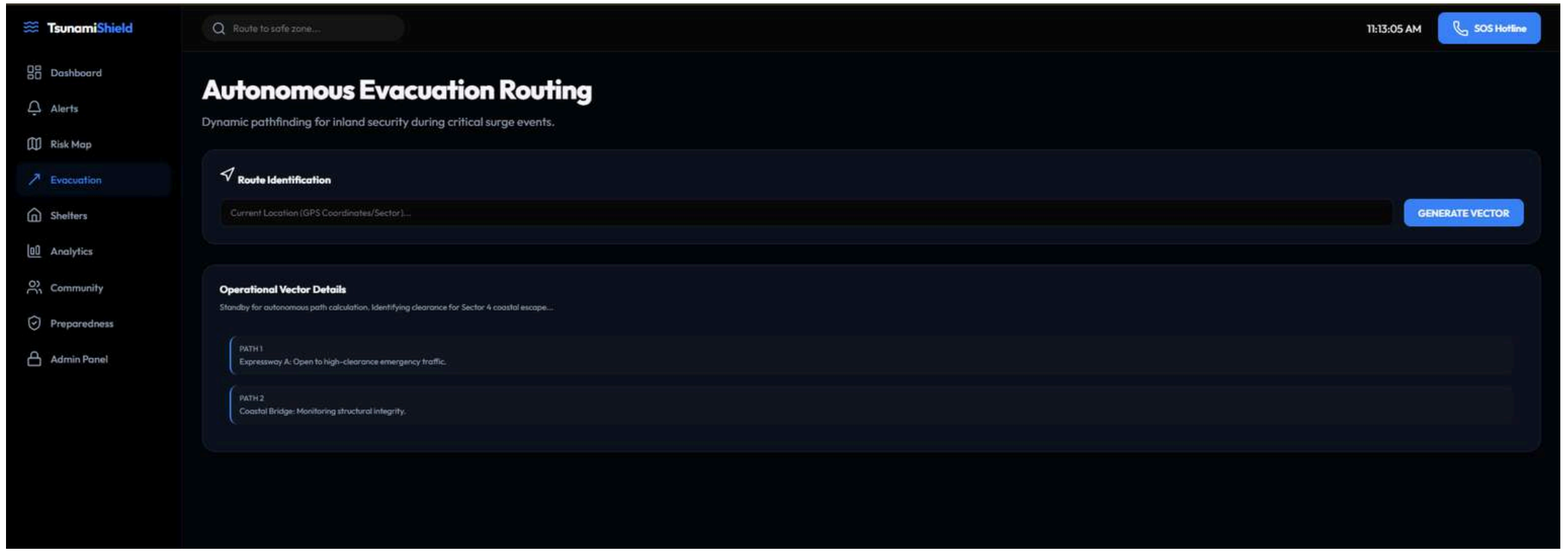
- **Real-time Shelter Tracking:** Displays nearby shelters with live capacity status (Available, Full, Filling Fast) to help users quickly identify safe locations.
- **Smart Evacuation Support:** Integrates map-based visualization with shelter data to guide users toward the nearest and most suitable evacuation center efficiently.

*This module provides a **real-time geospatial view of tsunami risk zones**, helping users and authorities quickly understand affected areas. It dynamically displays **safe zones** (green) and **high-risk zones** (red) on an interactive map, enabling faster decision-making and efficient evacuation planning.*



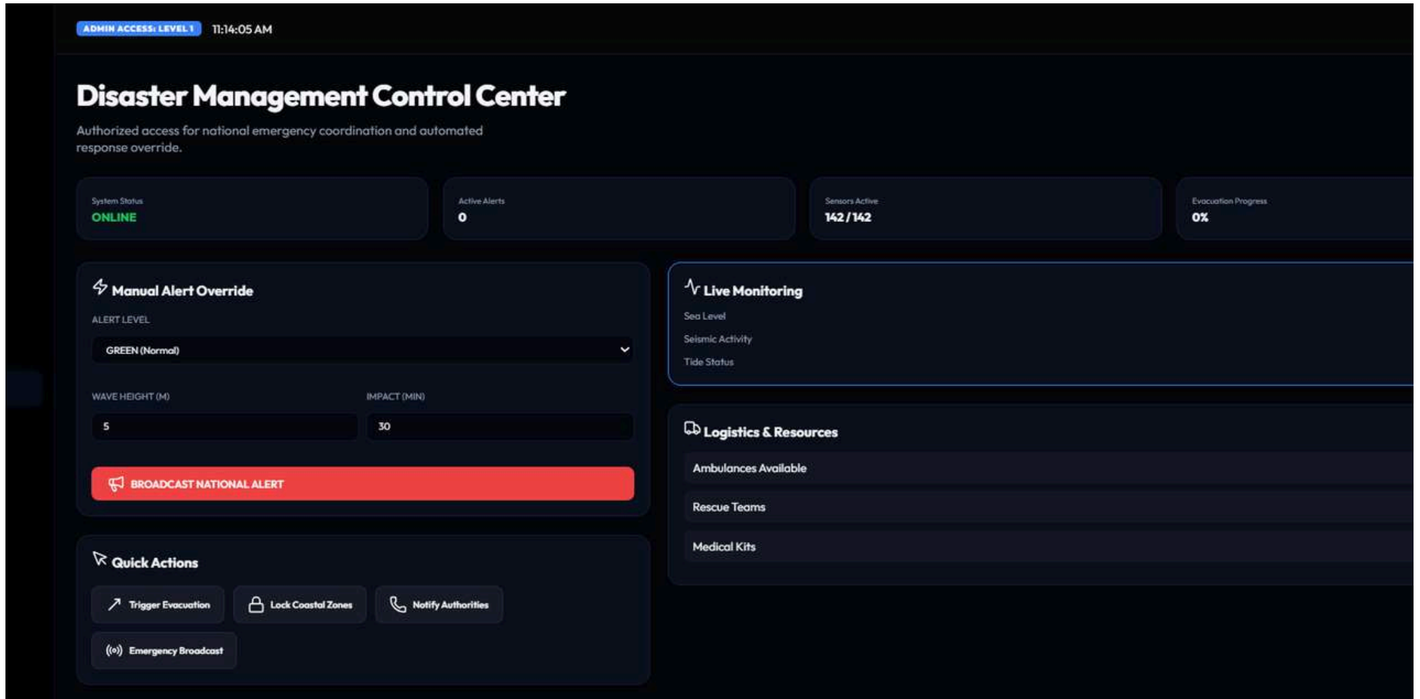
Tsunami

Precision Evacuation & Life Safety



*A centralized dashboard for authorities to **monitor**, **control**, and **respond** to disasters in **real-time**. It enables manual **alert triggering**, resource management, and live system monitoring, ensuring quick decision-making and coordinated emergency response.*

*This module provides **AI-driven evacuation route** planning, guiding users to the **safest paths** during emergencies. It analyzes **real-time conditions** to suggest **optimal** escape routes, ensuring **faster evacuation** while avoiding high-risk zones.*



FEASIBILITY AND VIABILITY

QUANTUM ARENA 1.0

Feasibility Analysis

Technical Feasibility

- Machine Learning model for fire prediction
- Quantum optimization for evacuation routing and resource allocation

Operational Feasibility

- Can be used by forest departments & disaster authorities
- AI agent automatically sends SMS alerts to nearby citizens during emergencies

Economic Feasibility

Cloud hosting reduces infrastructure cost

Potential Challenges & Risks

Data

Incomplete or noisy environmental data

Complexity

Dynamic environmental conditions

Quantum

Remote forest areas may have weak internet

Adoption

Training required for forest officers

Strategies to Overcome Challenges

Data

Data preprocessing & cleaning

Scale

Cloud-based architecture

Quantum

Hybrid classical + quantum optimization (QAOA) for evacuation routes and resource deployment.

Adoption

Training sessions & awareness programs

TARGET AUDIENCE

- Forest Department Officials
- Disaster Management Authorities
- Environmental Agencies
- Local Communities & Villagers
- Government & Climate Policy Makers

Subscription Model

Free

Forest Monitoring

- Fire Risk Prediction
- Alert Notifications
- Dashboard Monitoring

Basic

For District Authorities

- Safe Evacuation Route Mapping
- Cloud Storage
- Resource Optimization Engine
- Environmental Impact Estimation
- Automated Citizen SMS Alert System

Pro

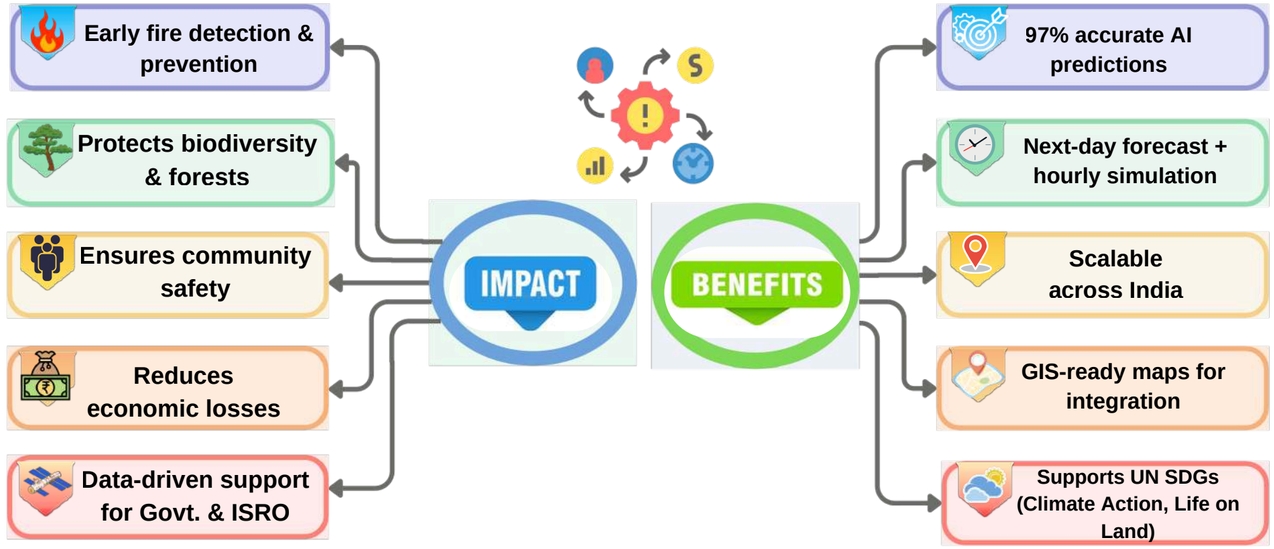
For Government

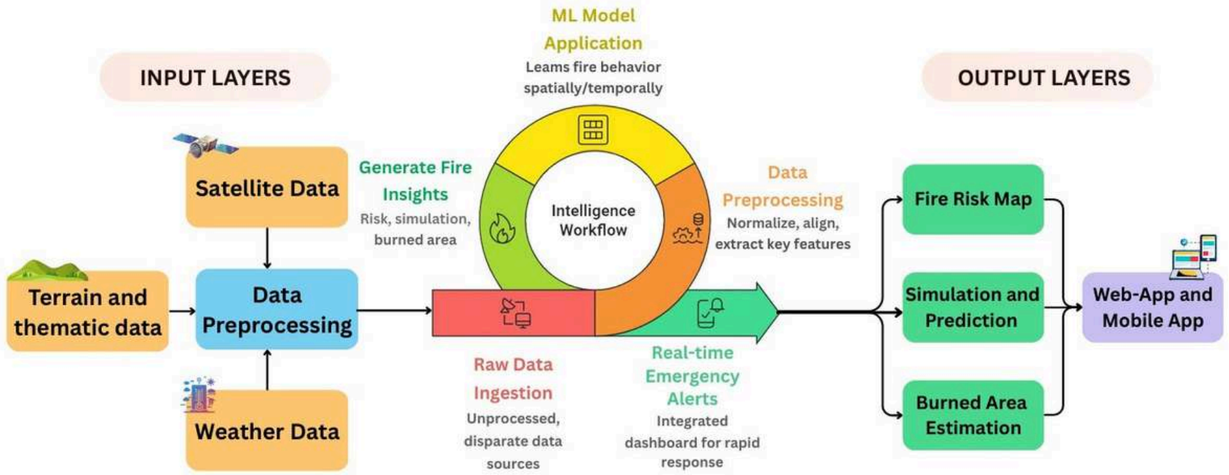
Basic +

- AI Explainability Module
- Crowdsourced Fire Reporting
- 3D Fire Spread Visualization
- Post-Fire Recovery Planning
- Quantum Resource Dispatch Optimization

IMPACT AND BENEFITS

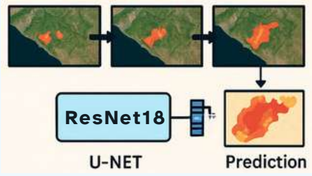
QUANTUM ARENA 1.0





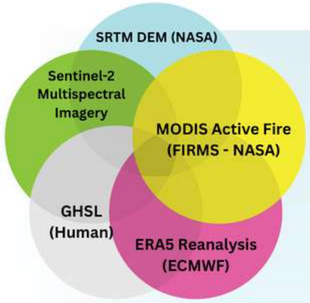
Forest Fire Intelligence Workflow

1.ConvLSTM + U-Net Hybrid Model



- ConvLSTM captures temporal features.
- U-Net segments **fire-prone regions**
- Combined : precise **next-day fire risk map**
- Optimized for **30m** resolution prediction

Smart prediction. Maximum accuracy.

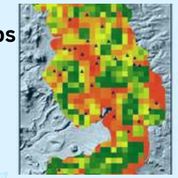


2. Unified Multi-Source Data Pipeline

List of features offered by the solution

3. Next Day Fire Risk Mapping

- Outputs **high resolution probability maps**
- Provides detailed **30m pixel** predictions
- Forecasts 1 day in advance using geospatial AI



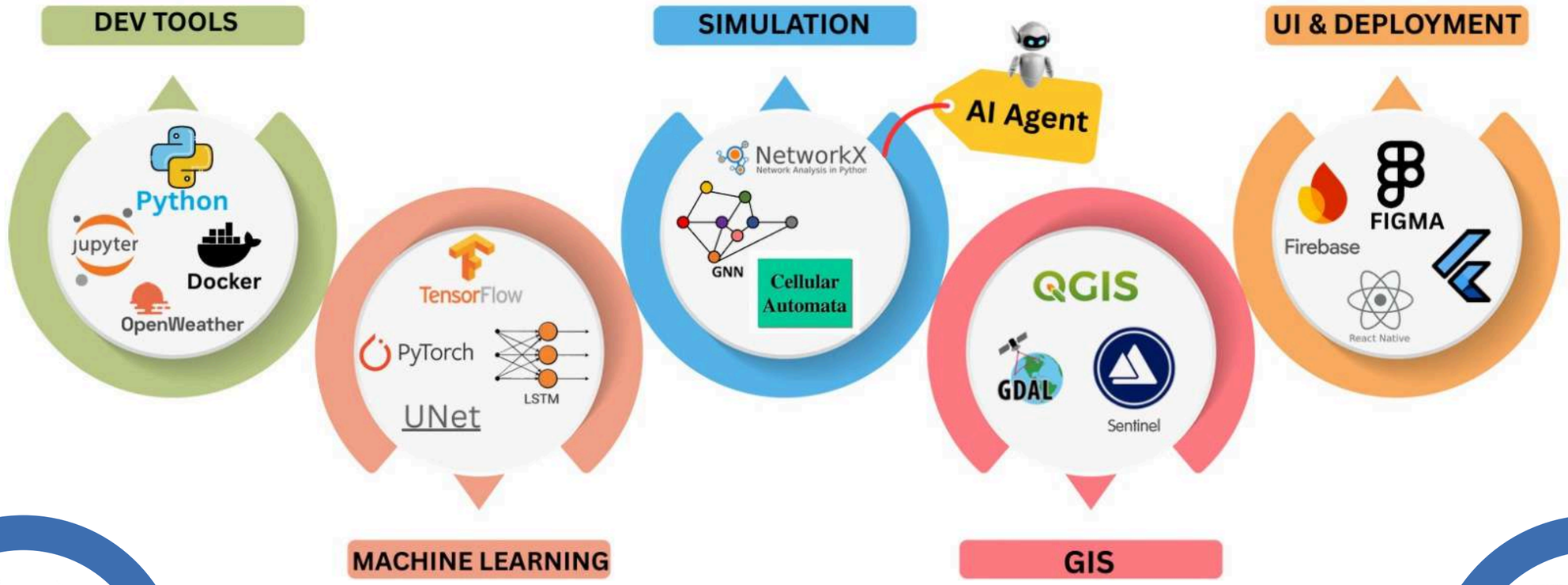
RESEARCH AND REFERENCES

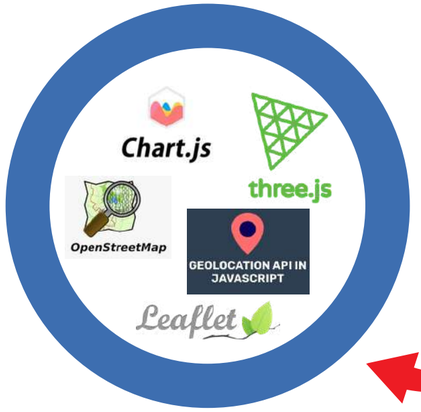
QUANTUM ARENA 1.0

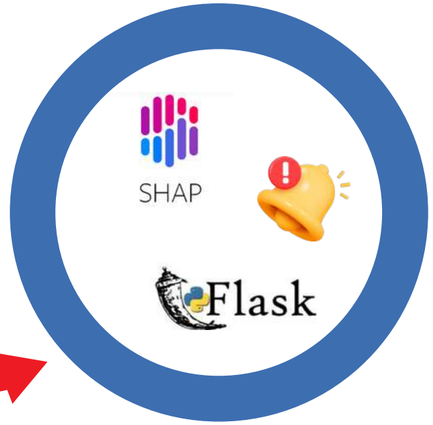
- **ConvLSTM** for Spatio-Temporal Prediction – Shi et al., NeurIPS 2015
- **U-Net** Segmentation Model – Ronneberger et al., MICCAI 2015
- CA-based Fire Spread **Simulation** – Alexandridis et al., Applied Math & Computation 2008
- **MODIS** Active Fire (NASA FIRMS) – Giglio et al., Remote Sensing of Environment 2016
- Sentinel-2 **Multispectral** Imagery – Copernicus Open Access Hub
- **ERA5** Weather Reanalysis Data – ECMWF
- SRTM Digital Elevation Model – **NASA Earth Data**
- Global Human Settlement Layer (GHSL) – EU JRC
- **NDVI & BAI** Index for Vegetation Monitoring – Chuvieco et al., Remote Sensing 2019
- **Global** Wildfire Trends & Impacts – Jain et al., Environmental Research Letters 2020

REQUIREMENTS

QUANTUM ARENA 1.0





DATA & APIs

AI EXPLAINABILITY